

Disorder in Cavity-Coupled Jahn–Teller Active Molecules

From single-molecule polaritons to disorder-averaged collective spectra

Summer Internship 2026

JT-Absorption-Spectra

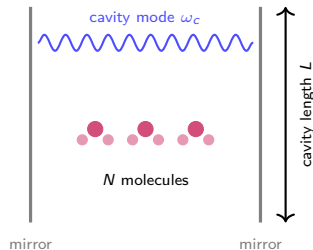
July 24, 2026

Outline

- 1 Cavity–matter interaction
- 2 Jahn–Teller active molecules
- 3 Single molecule: cavity Jahn–Teller polaritons (2023)
- 4 Collective vibronic cascade (2025)
- 5 Introducing disorder
- 6 Disorder in our model & results

Molecules in an optical cavity: polaritons

- A Fabry–Pérot cavity confines the electromagnetic field to discrete modes of frequency ω_c .
- When the light–matter coupling exceeds all loss rates (*strong coupling*), molecule and photon form hybrid eigenstates: **polaritons**.
- These split into a **lower** (LP) and **upper** (UP) polariton, separated by the **Rabi splitting** Ω .
- Coupling N molecules collectively enhances the interaction as $\sqrt{N}$: the single-molecule coupling is $g = \Omega/(2\sqrt{N})$.
- Polariton chemistry: modifying photophysics/reactivity by placing molecules in a cavity.



Collective strong coupling:
LP/UP split by $\Omega = 2\sqrt{N}g$.

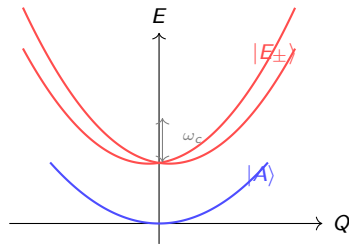
[Ref] T. W. Ebbesen, *Hybrid light–matter states*. . . , Acc. Chem. Res. 49, 2403 (2016). Model & figures: S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025).

The Jahn–Teller effect and the $(E \times e)$ model

Jahn–Teller theorem. A non-linear molecule in an orbitally degenerate electronic state spontaneously distorts to lift the degeneracy.

$(E \times e)$ **problem.** A doubly-degenerate electronic state E (components $|E_x\rangle, |E_y\rangle$) coupled to a doubly-degenerate vibrational mode e (coordinates Q_x, Q_y):

- ground state $|A\rangle$, excited doublet $|E_{\pm}\rangle = (|E_x\rangle \pm i|E_y\rangle)/\sqrt{2}$;
- potential surfaces form a *Mexican hat* with a **conical intersection**;
- vibronic coupling κ mixes electronic and nuclear motion $\Rightarrow$ **vibronic angular momentum** is exchanged.



Excited $(E \times e)$ surfaces (red) and ground $|A\rangle$ (blue)
vs. mode coordinate Q .

[Ref] H. A. Jahn & E. Teller, Proc. R. Soc. A 161, 220 (1937); $(E \times e)$: H. C. Longuet-Higgins *et al.*, Proc. R. Soc. A 244, 1 (1958). Hamiltonian: K. R. Umashankar & Q. Li, *Measuring cavity Jahn–Teller polaron in molecules*, Phys. Rev. A 107, L061101 (2023).

Cavity Jahn–Teller Hamiltonian — one molecule

Two perpendicular normal-incidence cavity modes (x, y polarized), recast as circular modes $\hat{a}_{\pm}$ with angular momentum $\pm\hbar$. The cavity–JT Hamiltonian is

$$\hat{H} = \hat{H}_{\text{JT}} + \hat{H}_C + \hat{H}_{CM}, \quad \hat{H}_C = \omega_c (\hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-),$$

$$\hat{H}_{CM} = \frac{\Omega}{2} (\hat{a}_+^\dagger |A\rangle \langle E_+| + \hat{a}_-^\dagger |A\rangle \langle E_-| + \text{h.c.}).$$

Conserved angular momentum

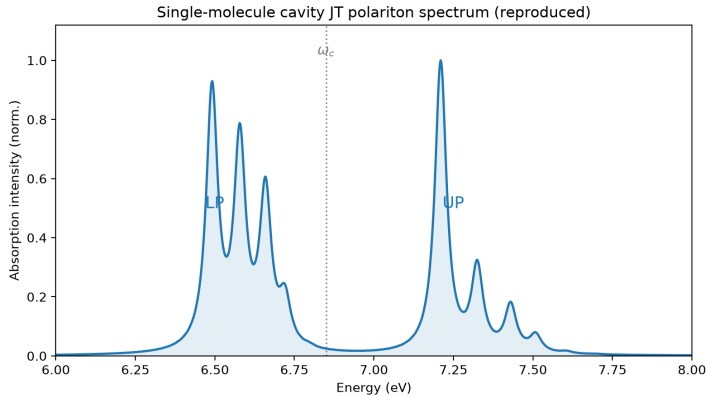
$$\hat{J} = 2\hat{L}_c + \hat{S}_z + \hat{l}_c \equiv 2L + s + l$$

- L_c : vibrational pseudorotation, S_z : electronic, l_c : photon helicity.
- Vibronic coupling $\Rightarrow$ efficient *exchange* of angular momentum between light, electrons and vibrations.

- Polaritons carry **mixed** (+/−) **circular polarization**, set by the molecular vibronic coupling.
- Net cavity polarization is strongly *suppressed* and becomes **energy dependent**.
- Some eigenstates show **inverted polarization** (opposite to the exciting field).

[Ref] K. R. Nandipati & O. Vendrell, *Cavity Jahn–Teller polaritons in molecules*, Phys. Rev. A **107**, L061101 (2023).

Single-molecule polaritonic spectrum (reproduced)



- Cavity set resonant with the brightest bare JT transition, $\omega_c = 6.85$ eV.
- Spectrum splits into **LP** and **UP** branches, each a few well-resolved vibronic peaks.
- For $N = 1$ only **3 vibronic sectors** $\nu \in \{0, -1, -2\}$ are accessible (bounded).

Parameters: $\epsilon = 7$ eV, $\omega = 0.082$ eV,
 $\kappa/\omega = 2.2$, $\Omega/2\omega_c = 0.05$.

[Ref] Physics & parameters: K. R. Nandipati & O. Vendrell, *Cavity Jahn–Teller polaritons in molecules*, Phys. Rev. A 107, L061101 (2023). Spectrum reproduced with our code.

N molecules in a common cavity

N identical JT molecules coupled to the two cavity modes:

$$\hat{H} = \hat{H}_c + \sum_{k=1}^N (\hat{H}_{m,k} + \hat{H}_{m-c,k}), \quad \hat{H}_{m-c,k} = \frac{\Omega}{2\sqrt{N}} (\hat{a}_+^\dagger |A\rangle \langle E_+|_k + \hat{a}_-^\dagger |A\rangle \langle E_-|_k + \text{h.c.})$$

The $\sqrt{N}$ normalization keeps the collective coupling $g = \Omega/(2\sqrt{N})$ physical. Two conserved quantities:

Excitation number

$$\hat{N}_{\text{ex}} = \sum_k (|E_+\rangle \langle E_+|_k + |E_-\rangle \langle E_-|_k) + \hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-$$

Total angular momentum

$$\hat{J} = \sum_{k=1}^N \hat{V}_k + \hat{P}, \quad \hat{V}_k = 2\hat{L}_{z,k} + \hat{S}_{z,k}$$

We work in the optically-relevant single-excitation sector ($n_{\text{ex}} = 1$, $j = -1$).

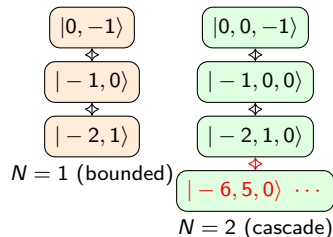
[Ref] S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025).

The collective vibronic cascade

Each molecule's JT Hamiltonian has vibronic eigenstates with a definite **vibronic angular momentum** $\nu = 2(n_+ - n_-) + S_z$:

$$\hat{H}_{m,k} = \sum_{\nu} \sum_i \lambda_i^{\nu} |\lambda_i^{\nu}\rangle \langle \lambda_i^{\nu}|_k$$

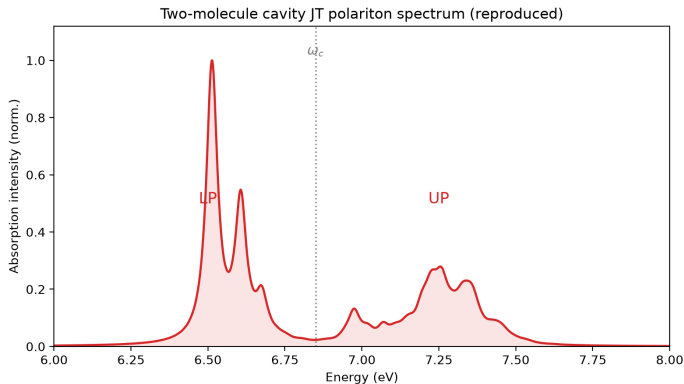
- $N = 1$: bounded — only $\nu \in \{0, -1, -2\}$ reachable $\Rightarrow$ participation ratio $\text{PR} \leq 3$.
- $N \geq 2$: the cavity photon shuttles angular momentum *between* molecules $\Rightarrow$ an unbounded ladder of ν sectors — the **collective vibronic cascade**.



Cascade metric: $\text{PR} = 1/\sum P(\nu)^2$.

[Ref] S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025).

Two-molecule polaritonic spectrum (reproduced)

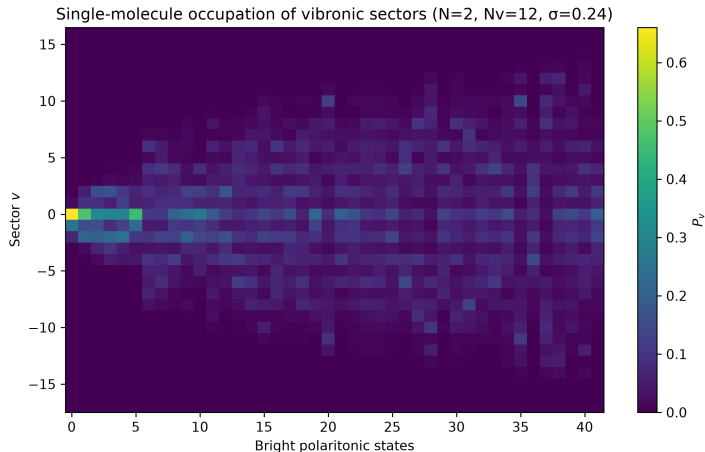


- LP branch stays fairly sharp; **UP branch broadens** into many short stems — the cascade fingerprint.
- $N = 2$ reaches PR up to ~ 20 (vs ≤ 3 for $N = 1$).
- Broadening grows with N ; photon-polarization dynamics get suppressed.

We reproduce the $N = 2$ spectrum and the vibronic-sector heatmap (next section) of the paper.

[Ref] Physics: S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025). Spectrum reproduced with our code.

Vibronic-sector heatmap $P(v)$ — clean $N = 2$ (reproduced)



For every *bright* polaritonic eigenstate we plot the probability $P(v)$ that a molecule occupies vibronic sector v :

$$P^{(k)}(v_0) = \sum_{a: v_a=v_0} \sum_{b,p} |C_{a,b,p}|^2$$

- x: bright states (energy-ordered); y: sector v ; color: $P(v)$.
- Clean $N = 2$: **42 bright states**, $\text{PR}_{\text{max}} \approx 20$, $v \in [-15, 14]$.
- This is the paper's Fig. S1 — our reference baseline.

[Ref] Fig. S1 of S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025). Reproduced with our code.

Why disorder? Two guiding papers

Real molecular ensembles are **inhomogeneous**: each molecule has a slightly different transition energy (static *energy disorder*). Two works frame what disorder does to a cavity-coupled ensemble:

(a) Arrowhead-matrix picture (single excitation)

In the single-excitation sector the disordered Tavis–Cummings Hamiltonian is an **arrowhead matrix**: a disordered diagonal of molecular energies plus one coupling row/column to the photon. Its eigenstates are two **polariton outliers** (split by $\pm\sqrt{N}g$) plus $N - 1$ **dark states**; disorder induces *semi-localization* and reshapes the photon spectral function.

(b) Disordered Holstein–Tavis–Cummings (with vibrations)

Adds vibrations (Holstein coupling). Central result: **disorder enhances** excitation transfer and vibrational entanglement, and dark states *acquire photon weight* (brighten). Effect peaks when disorder $W \sim g_c$.

[Ref] (a) C. Dubail, T. Botzung, J. Schachenmayer, G. Pupillo, D. Hagenmüller, *Large Random Arrowhead Matrices...*, arXiv:2105.08444 (2021). (b) D. Wellnitz, G. Pupillo, J. Schachenmayer, *Disorder enhanced vibrational entanglement and dynamics in polaritonic chemistry*, Commun. Phys. 5, 120 (2022).

Disorder: the equations

Disordered Holstein–Tavis–Cummings model (Wellnitz *et al.*):

$$\hat{H} = \hat{H}_{\text{TC}} + \hat{H}_{\text{vib}} + \hat{H}_{\text{H}} + \hat{H}_{\text{dis}},$$

$$\hat{H}_{\text{dis}} = \sum_n \epsilon_n \hat{\sigma}_n^+ \hat{\sigma}_n^-$$

with i.i.d. energy offsets ϵ_n of mean 0 and variance W^2 (Gaussian). Consequences quantified in the papers:

- Dark states gain photon weight
 $\sum_{\text{dark}} |\langle d | 1_{\text{ph}} \rangle|^2 \approx W^2 / 2g_c^2 \Rightarrow$ they **appear in absorption**.
- Polaritons stay protected while g beats disorder: crossover at $W \sim 2g$.
- Arrowhead: photon spectral function
 $A(\epsilon) = \sum_a \text{PW}_a \delta(\epsilon - \epsilon_a)$ — this *is* the absorption spectrum.
- Eigenstate amplitudes $\psi_{a,j} \sim 1/(\epsilon_a - \omega_j)^2$; level statistics Poisson $\rightarrow$ semi-Poisson.

Caveat

These results are for the large- N ($N \rightarrow \infty$) limit. We study the $N = 2$ **analog** — so we use them as the conceptual framework, not as claims of multifractality at $N = 2$.

[Ref.] P. Wellnitz, G. Pupillo, J. Schachenmayer, *Disorder enhanced vibrational entanglement and dynamics in polaritonic chemistry*, *Commun. Phys.* **5**, 129 (2022). (HTC, $\hat{H}_{\text{dis}}$, $W^2/2g_c^2$); C. Dubail, T. Botzung, J. Schachenmayer, G. Pupillo, D. Hagenmüller, *Large Random Arrowhead Matrices...*, *arXiv:2105.08444 [cond-mat.str-el]* (April Statistics)

How disorder enters our model

We add **Gaussian site-energy disorder** to the two JT molecules' transition energies:

$$\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2), \quad i = 1, 2, \quad \epsilon = 7.0 \text{ eV}.$$

Per disorder *realization* we build $\hat{H}$ in the ($n_{\text{ex}} = 1, j = -1$) sector, diagonalize, take the intensity of each eigenstate as $|\langle i_0 | \psi \rangle|^2$ (overlap with the ground state + one photon), broaden with a Lorentzian (FWHM $\gamma = 0.044$ eV), and **average over realizations**.

Energy scales (from spectrum/config.py)

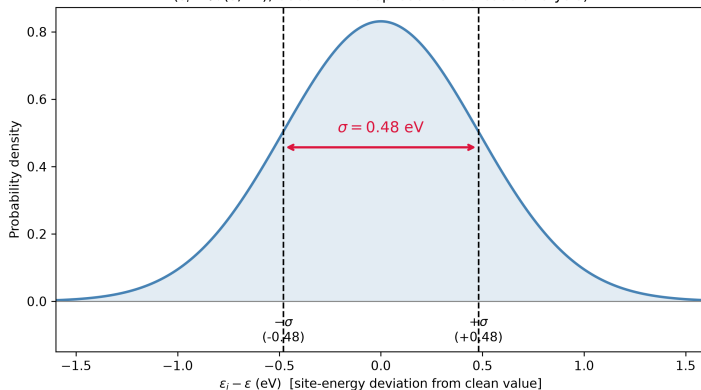
vibrational quantum ω	0.082 eV	JT coupling $\kappa/\omega = 2.2$
per-molecule coupling $g = \Omega/2\sqrt{N}$	0.242 eV	cavity $\omega_c = 6.85$ eV
Rabi splitting $\Omega = 2\sqrt{N}g$	0.685 eV	sites $\epsilon = 7.0$ eV, $N = 2$

Disorder strength quoted relative to g : we run at $\sigma \approx 2g$ (strong-disorder band), where the representative cases are physically distinct.

This work — computed with our code (JT-Absorption-Spectra).

Disorder realizations: the Gaussian distribution

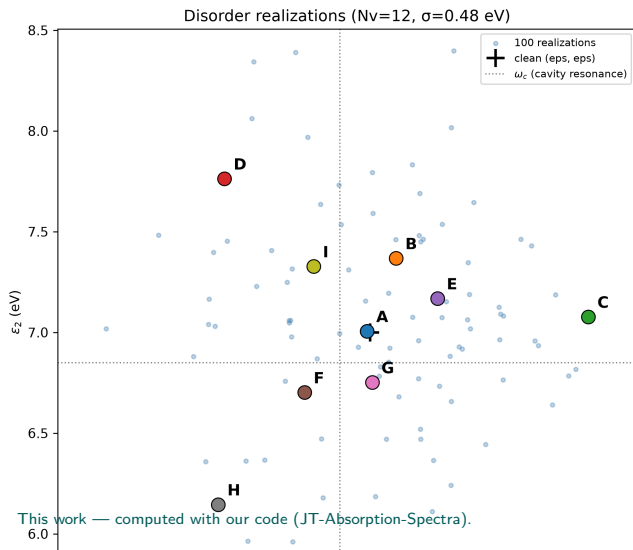
Gaussian disorder distribution, $\sigma = 0.48$ eV
($\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2)$, used in the representative-case analysis)



- Each molecule's site energy is drawn from $\mathcal{N}(\epsilon, \sigma^2)$ with $\sigma = 0.48$ eV.
- $\sigma/g \approx 2.0$, $\sigma/\Omega \approx 0.70$ — just below the $W = 2g$ crossover.
- Dashed lines / arrow mark $\pm\sigma$ (-0.48 to $+0.48$ eV).
- We average absorption over 100 such realizations.

[Ref] Disorder-strength choice motivated by D. Wellnitz, G. Pupillo, J. Schachenmayer, *Disorder enhanced vibrational entanglement and dynamics in polaritonic chemistry*, Commun. Phys. **5**, 120 (2022). / C. Dubail, T. Botzung, J. Schachenmayer, G. Pupillo, D. Hagenmüller, *Large Random Arrowhead Matrices...*, arXiv:2105.08444 (2021). Plot: our code

Representative-realization analysis: the (ϵ_1, ϵ_2) cloud



Behind the disorder average lie many individual (ϵ_1, ϵ_2) draws. We pick **9 physically meaningful realizations** (Cases A–I), each best satisfying a criterion on the actual drawn energies:

- common-mode shift $s = (\delta_1 + \delta_2)/2$ vs. $\sqrt{N}g$;
- inter-molecular mismatch $\Delta = \epsilon_1 - \epsilon_2$ vs. g .

Dotted lines mark ω_c ; the “+” marks the clean (ϵ, ϵ) . Each case is then recomputed individually.

The nine representative cases ($\sigma = 0.48$ eV)

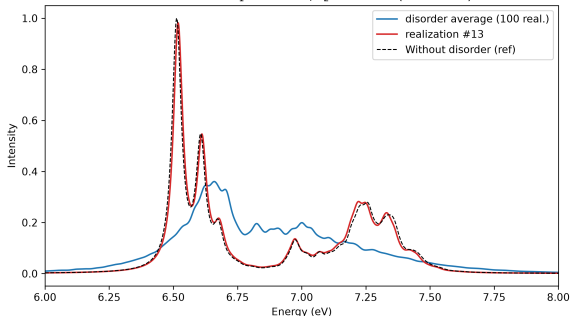
Case	Description	(ϵ_1, ϵ_2)	Δ	#bright	verdict
A	Resonant baseline (no disorder)	(6.98, 7.01)	0.02	39	$\approx$ clean
B	Mismatch $\approx$ coupling	(7.13, 7.37)	g	14	partial
C	Large mismatch (localized)	(8.08, 7.08)	$4g$	11	dead
D	Opposite disorder	(6.28, 7.76)	$6g$	14	partial $\rightarrow$ dead
E	Common blue detuning	(7.33, 7.17)	0.17	12	partial
F	Common red (matched)	(6.67, 6.70)	0.03	104	enhanced
G	Single-molecule resonant	(7.01, 6.75)	g	132	enhanced
H	Bare-molecule limit	(6.25, 6.15)	0.10	8	dead
I	A–D line, closest to ω_c	(6.72, 7.33)	$2.5g$	46	intermediate

#bright = number of bright vibronic (polaritonic) states above 1% of the peak. Clean $N = 2$ reference: **42**. Disorder can *brighten* dark states (F, G, I $> 42/39$) or kill the cascade (C, H).

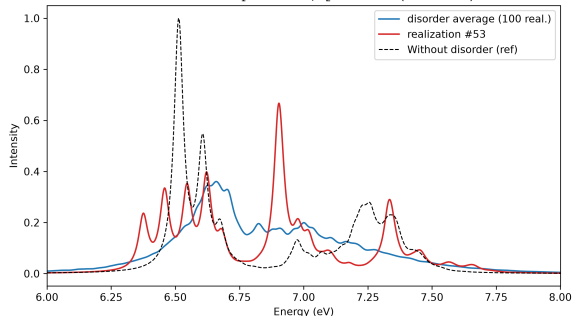
This work — computed with our code (JT-Absorption-Spectra).

Case spectra: baseline vs. A-D line

A: Resonant baseline
realization #13: $\varepsilon_1=6.984$ eV, $\varepsilon_2=7.006$ eV ($\sigma=0.48$ eV)



I: A-D line, closest to ω_c
realization #53: $\varepsilon_1=6.720$ eV, $\varepsilon_2=7.328$ eV ($\sigma=0.48$ eV)



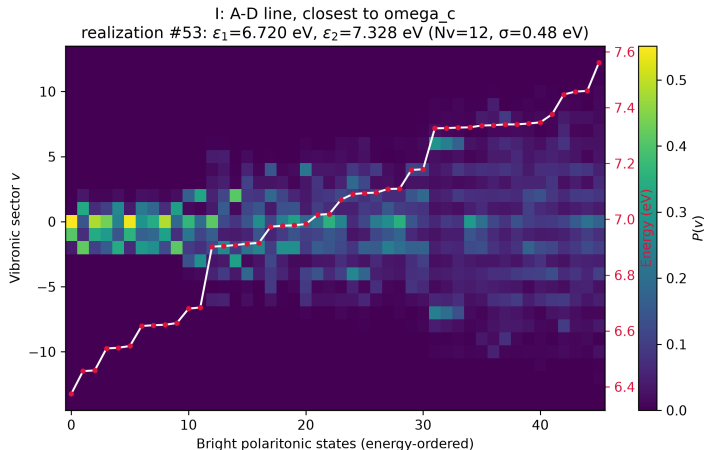
Case A — near-zero disorder reproduces the clean collective cascade.

Cascade weakens monotonically along the A-D line: PR 20.6 (A) $\rightarrow$ 16.2 (I) $\rightarrow$ 8.5 (D) as opposite-disorder mismatch grows.

This work — computed with our code (JT-Absorption-Spectra).

Case I — interior point on the A-D line: broad, spread oscillator strength, brightest near ω_c .

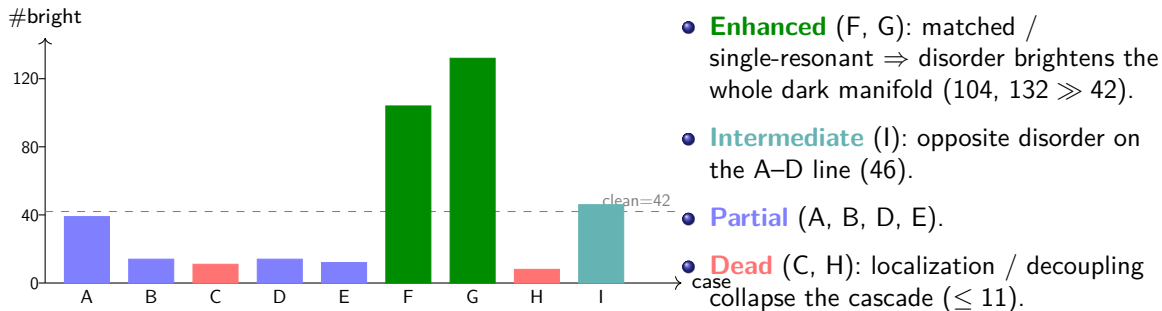
Heatmap analysis with LP/UP energy axis



- Left y-axis: vibronic sector v ; color: $P(v)$.
- Right y-axis (red): each bright state's **energy** vs. the same index — a *jump* marks the LP $\rightarrow$ UP branch boundary.
- Case I: **46 bright states**, $PR_{\max} = 16.2$, $v \in [-14, 13]$.
- Opposite disorder *brightens* dark states ($46 > 39$ at A) yet lowers per-state spread.

[Ref] $P(v)$ heatmap def. from Fig. S1 of S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025).. Energy-axis overlay & computation: our code.

Number of bright vibronic states vs. case



This work — computed with our code (JT-Absorption-Spectra).

Summary

- 1 **Reproduced** the single-molecule cavity-JT polariton spectrum (LP/UP, bounded ν) and the two-molecule collective spectrum + $P(\nu)$ heatmap (unbounded cascade, PR ~ 20).
- 2 **Introduced disorder** following the arrowhead-matrix and disordered Holstein–Tavis–Cummings frameworks: Gaussian site-energy disorder $\epsilon_i \sim \mathcal{N}(\epsilon, \sigma^2)$, dark states brighten, cascade protected while g beats σ .
- 3 **Implemented & analyzed** disorder in our $N = 2$ model: realizations, representative cases (A–I), and $P(\nu)$ heatmaps with an LP/UP energy overlay.
- 4 **Key result:** the number of bright vibronic states and the cascade PR track the two disorder axes — *enhanced* (F, G, I), *partial*, or *dead* (C, H).

Outlook: larger N , LP/UP-resolved PR correlation, stronger disorder ($\sigma \sim 10g$) to test cascade survival.

- ❶ K. R. Nandipati & O. Vendrell, *Cavity Jahn–Teller polaritons in molecules*, Phys. Rev. A **107**, L061101 (2023).
- ❷ S. K. Pandit, A. Pandey, A. Shankar, K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn–Teller Active Molecules*, arXiv:2511.07880 (2025).
- ❸ C. Dubail, T. Botzung, J. Schachenmayer, G. Pupillo, D. Hagenmüller, *Large Random Arrowhead Matrices. . .*, arXiv:2105.08444 (2021).
- ❹ D. Wellnitz, G. Pupillo, J. Schachenmayer, *Disorder enhanced vibrational entanglement and dynamics in polaritonic chemistry*, Commun. Phys. **5**, 120 (2022).
- ❺ H. A. Jahn & E. Teller, Proc. R. Soc. A **161**, 220 (1937); (E×e): H. C. Longuet-Higgins et al., Proc. R. Soc. A **244**, 1 (1958).
- ❻ T. W. Ebbesen, *Hybrid light–matter states. . .*, Acc. Chem. Res. **49**, 2403 (2016).